

Answer ALL questions. Write your answers in the spaces provided.

1. A researcher is investigating the distribution of orchids in a field. He believes that the Poisson distribution with a mean of 1.75 may be a good model for the number of orchids in each square metre. He randomly selects 150 non-overlapping areas, each of one square metre, and counts the number of orchids present in each square.

The results are recorded in the table below.

Number of orchids in each square metre	0	1	2	3	4	5	6
Number of squares	30	42	35	26	11	6	0

He calculates the **expected** frequencies as follows

Number of orchids in each square metre	0	1	2	3	4	5	More than 5
Number of squares	26.07	45.62	39.91	23.28	10.19	3.57	r

- (a) Find the value of r giving your answer to 2 decimal places.

(1)

The researcher will test, at the 5% level of significance, whether or not the data can be modelled by a Poisson distribution with mean 1.75

- (b) State clearly the hypotheses required to test whether or not this Poisson distribution is a suitable model for these data.

(1)

The test statistic for this test is 2.0 and the number of degrees of freedom to be used is 4

- (c) Explain fully why there are 4 degrees of freedom.

(2)

- (d) Stating your critical value clearly, determine whether or not these data support the researcher's belief.

(2)

The researcher works in another field where the number of orchids in each square metre is known to have a Poisson distribution with mean 1.5

He randomly selects 200 non-overlapping areas, each of one square metre, in this second field, and counts the number of orchids present in each square.

- (e) Using a Poisson approximation, show that the probability that he finds at least one square with exactly 6 orchids in it is 0.506 to 3 decimal places.

(4)



1. A leisure club offers a choice of one of three activities to its 150 members on a Tuesday evening. The manager believes that there may be an association between the choice of activity and the age of the member and collected the following data.

Activity Age a years	Badminton	Bowls	Snooker
$a < 20$	9	3	3
$20 \leq a < 40$	10	10	14
$40 \leq a < 50$	16	15	5
$50 \leq a < 60$	15	13	11
$a \geq 60$	4	19	3

- (a) Write down suitable hypotheses for a test of the manager's belief.

(1)

The manager calculated expected frequencies to use in the test.

- (b) Calculate the expected frequency of members aged 60 or over who choose snooker, used by the manager.

(1)

- (c) Explain why there are 6 degrees of freedom used in this test.

(2)

The test statistic used to test the manager's belief is 19.583

- (d) Using a 5% level of significance, complete the test of the manager's belief.

(2)

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2. A spinner used for a game is designed to give scores with the following probabilities

Score	1	2	3	4	6
Probability	$\frac{3}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{5}$	$\frac{1}{10}$

The spinner is spun 80 times and the results are as follows

Score	1	2	3	4	6
Frequency	15	4	12	41	8

Test, at the 10% level of significance, whether or not the spinner is giving scores as it is designed to do. Show your working and state your hypotheses clearly.

(7)

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4. Charlie carried out a survey on the main type of investment people have.
The contingency table below shows the results of a survey of a random sample of people.

		Main type of investment		
		Bonds	Cash	Stocks
Age	25–44	a	$b - e$	e
	45–75	c	$d - 59$	59

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- (a) Find an expression, in terms of a , b , c and d , for the difference between the observed and the expected value ($O - E$) for the group whose main type of investment is Bonds and are aged 45–75

Express your answer as a single fraction in its simplest form.

(4)

Given that $\sum \frac{(O - E)^2}{E} = 9.62$ for this information,

- (b) test, at the 5% level of significance, whether or not there is evidence of an association between the age of a person and the main type of investment they have. You should state your hypotheses, critical value and conclusion clearly.

You may assume that no cells need to be combined.

(3)



- (1)**

Number of heads	0	1	2	3	4	5
Observed frequency	2	27	93	181	146	51

Number of heads	0	1	2	3	4	5
Expected frequency	5.12	38.40	115.20	172.80	129.60	38.88

- (b) State the number of degrees of freedom in Chris' test, giving a reason for your answer. (1)
- (c) Carry out the test at the 5% level of significance. (5)
You should state your hypotheses, test statistic, critical value and conclusion clearly.
- (d) Showing your working, find an alternative model which would better fit Chris' data. (2)

2. A bag contains a large number of balls, all of the same size and weight. The balls are coloured Red, Blue or Yellow.

Jasmine asks each child in a group of 150 children to close their eyes, select a ball from the bag and show it to her. The child then replaces the ball and repeats the process a second time.

If both balls are the same colour the child receives a prize.

The results are given in the table below.

1st colour 2nd colour	Red	Blue	Yellow	Total
Red	31	11	18	60
Blue	8	10	9	27
Yellow	21	9	33	63
Total	60	30	60	150

Jasmine carries out a test, at the 5% level of significance, to see whether or not the colour of the 2nd ball is independent of the colour of the 1st ball.

- (a) Calculate the expected frequencies for the cases where both balls are the same colour.

(2)

The test statistic Jasmine obtained was 12.712 to three decimal places.

- (b) Use this value to complete the test, stating the critical value and conclusion clearly.

(3)

With reference to your calculations in part (a) and the nature of the experiment,

- (c) give a plausible reason why Jasmine may have obtained her conclusion in part (b).

(1)



4. Table 1 below shows the number of car breakdowns in the *Snoreap* district in each of 60 months.

Number of car breakdowns	0	1	2	3	4	5
Frequency	12	11	19	14	3	1

Table 1

Anja believes that the number of car breakdowns per month in *Snoreap* can be modelled by a Poisson distribution. Table 2 below shows the results of some of her calculations.

Number of car breakdowns	0	1	2	3	4	≥ 5
Observed frequency (O_i)	12	11	19	14	3	1
Expected frequency (E_i)	9.92			9.64	4.34	

Table 2

- (a) State suitable hypotheses for a test to investigate Anja's belief. (1)
- (b) Explain why Anja has changed the label of the final column to ≥ 5 (1)
- (c) Showing your working clearly, complete Table 2 (4)
- (d) Find the value of $\frac{(O_i - E_i)^2}{E_i}$ when the number of car breakdowns is
- (i) 1
- (ii) 3 (2)
- (e) Explain why Anja used 3 degrees of freedom for her test. (2)

The test statistic for Anja's test is 6.54 to 2 decimal places.

- (f) Stating the critical value and using a 5% level of significance, complete Anja's test. (2)





- (2)

(1)

(2)

(6)